

PARENT, QC, Canada

WMO: 717260

Lat: 47.922N

Lon: 74.624W

Elev: 1459

StdP: 13.94

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.4	(c) -23.7	(d) -37.3	(e) 0.7	(f) -28.9	(g) -31.4	(h) 1.0	(i) -22.8	(j) 15.1	(k) 8.8	(l) 13.6	(m) 9.1	(n) 1.5	(o) 290	(p) 0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.2	(c) 82.0	(d) 66.3	(e) 78.8	(f) 65.2	(g) 75.8	(h) 63.6	(i) 70.3	(j) 77.5	(k) 68.3	(l) 74.9	(m) 66.4	(n) 72.6	(o) 5.6	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.9	108.4	73.5	66.0	101.2	71.7	64.0	94.2	69.8	35.2	78.0	33.5	75.0	31.9	72.8	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.8	(b) 11.3	(c) 9.8	(e) -37.2	(f) 87.6	(g) 6.3	(h) 2.6	(i) -41.8	(j) 89.4	(k) -45.5	(l) 90.9	(m) -49.0	(n) 92.3	(o) -53.6	(p) 94.2		
(4) 12.8	(5) 11.3	(6) 9.8	(7) -37.2	(8) 73.2	(9) 5.8	(10) 2.5	(11) -41.5	(12) 75.0	(13) -44.9	(14) 76.5	(15) -48.1	(16) 77.9	(17) -52.4	(18) 79.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.3	3.1	6.3	18.0	32.6	48.6	58.1	62.6	60.1	52.4	40.7	26.8	12.4		
	DBStd	23.16	14.77	13.32	13.44	9.75	8.72	7.63	5.77	5.99	7.94	8.53	10.34	12.95		
	HDD50	6581	1454	1222	993	524	133	15	0	3	63	311	697	1166		
	HDD65	10939	1919	1642	1458	971	512	231	117	173	384	754	1147	1631		
	CDD50	1220	0	0	0	3	90	259	392	316	137	23	0	0		
	CDD65	102	0	0	0	0	5	25	43	22	7	0	0	0		
	CDH74	1035	0	0	0	1	104	311	374	191	51	3	0	0		
	CDH80	184	0	0	0	0	21	68	65	23	7	0	0	0		
(14) Wind	WSAvg	3.9	4.2	4.3	4.6	4.7	4.2	3.4	3.1	2.9	3.1	4.2	4.5	4.0		
(15) Precipitation	PrecAvg	40.7	3.4	2.1	2.3	2.8	2.8	3.7	4.5	4.0	4.0	3.6	3.7	4.1		
	PrecMax	61.4	11.0	4.1	5.4	4.3	5.1	6.7	8.9	7.2	8.1	5.9	10.3	14.9		
	PrecMin	31.7	0.9	0.8	1.1	1.7	0.9	1.3	1.0	0.9	1.5	1.3	0.8	1.5		
	PrecStd	6.9	2.1	0.9	0.9	0.8	1.0	1.3	1.7	1.5	1.8	1.3	1.8	2.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.2	41.8	55.6	67.0	82.8	86.2	85.6	83.1	79.5	71.1	55.2	41.1		
		MCWB	42.8	39.1	44.2	52.6	61.4	68.3	69.0	68.4	67.4	60.4	51.8	39.8		
	2%	DB	35.1	35.0	46.0	58.7	76.5	81.6	81.4	79.0	74.3	63.8	48.5	35.9		
		MCWB	34.5	32.4	38.0	46.3	58.3	65.2	67.5	66.8	64.6	55.3	45.4	34.9		
	5%	DB	29.9	31.4	40.9	53.1	71.4	77.5	78.2	75.7	70.1	58.8	44.0	33.1		
		MCWB	28.7	29.6	34.7	43.6	55.4	62.9	66.5	65.1	62.5	53.2	41.6	32.2		
	10%	DB	25.3	27.7	37.0	48.0	66.0	73.2	75.0	72.3	66.6	54.3	39.9	30.5		
		MCWB	24.1	25.8	33.0	40.0	53.4	61.1	64.6	63.6	60.3	50.0	38.0	29.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.9	38.9	47.6	55.0	66.2	72.3	74.5	71.9	69.3	61.8	52.2	40.0		
		MCDB	43.4	39.8	53.1	64.9	75.5	80.9	79.6	79.6	76.0	68.4	54.0	40.5		
	2%	WB	34.8	32.3	39.2	49.0	61.9	68.7	70.9	69.0	66.8	57.4	46.3	35.3		
		MCDB	35.2	33.9	43.1	55.6	70.9	76.3	77.8	75.9	71.9	62.5	47.9	35.9		
	5%	WB	29.0	29.2	35.9	44.7	58.7	66.1	68.6	67.0	64.1	53.9	41.9	32.5		
		MCDB	29.9	31.0	40.0	50.9	67.1	73.2	74.9	73.0	69.0	58.0	43.5	33.1		
	10%	WB	24.1	25.8	32.8	41.0	55.2	63.6	66.5	64.8	61.1	50.2	38.1	29.7		
		MCDB	25.1	27.3	36.3	47.2	63.4	70.4	73.0	70.2	65.6	53.5	39.8	30.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.2	23.7	23.0	20.5	23.5	22.2	20.2	19.6	19.3	14.7	12.9	16.3		
		MCDBR	18.6	20.3	25.2	29.2	33.3	28.1	24.8	24.8	24.0	22.9	18.0	13.0		
	5% WB	MCWBR	18.5	18.5	18.3	18.4	16.7	13.7	13.3	13.6	14.8	15.6	15.4	12.7		
		MCWBR	18.9	19.2	21.3	25.3	25.9	22.7	20.5	20.8	20.2	20.4	16.4	12.0		
(40) Clear-Sky Solar Irradiance	taub	taub	0.269	0.279	0.298	0.377	0.345	0.360	0.378	0.359	0.353	0.319	0.292	0.261		
		taud	2.229	2.219	2.230	2.196	2.449	2.428	2.389	2.440	2.446	2.535	2.477	2.320		
	Ebn at Noon	Ebn at Noon	254	279	292	277	291	286	279	280	271	266	249	245		
		Edh at Noon	24	31	36	42	34	35	36	33	30	23	20	20		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	406	716	1157	1456	1618	1753	1647	1420	1105	627	341	281		
	RadStd	RadStd	59	108	110	113	146	127	174	92	112	79	49	29		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page